

Project Proposal

School Result Portal

Capstone Project Proposal

Prepared by: DevCore 7-Innovators

1. Project Title

School Result Portal: A Web-Based Academic Result Management System

2. Introduction

This capstone project aims to design and develop a responsive web application that simplifies result management for administrators, teachers, students, and parents while improving accuracy, accessibility, and efficiency.

3. Problem Statement

Many schools still prepare students' results manually using spreadsheets or paper records. This approach presents several challenges:

- *Time-consuming result computation
- *Human errors in grading and calculations
- *Difficulty storing and retrieving records
- *Delayed publication of results
- *Limited access for students and parents
- *Poor data security

4. Aim of the Project

To design and develop a secure web-based application that automates the management, computation, storage, and publication of students' academic results.

5. Objectives

The project aims to:

- *Develop a secure login system for different users.
- *Enable administrators to manage students, teachers, subjects, classes, sessions, and terms.
- *Allow teachers to upload Continuous Assessment (CA) and examination scores.
- *Automatically calculate total scores, grades, averages, and class positions.
- *Allow students to securely view and download their academic results.
- *Generate printable report cards.
- *Provide administrators with performance reports and analytics.

6. Scope of the Project

The system will include the following modules:

Administrator

- *User management
- *Student registration
- *Teacher management

- *Subject management
- *Class management
- *Academic session management
- *Result approval

Teacher

- *Login
- *Score entry
- *Edit scores
- *Submit results

Student

- *Secure login
- *View results
- *Download report cards
- *View academic history

Parent (Optional)

- *Monitor student's academic performance

7. Expected Features

- *Secure Authentication
- *Role-Based Access Control
- *Student Information Management
- *Result Upload

- *Automatic Grade Calculation
- *Automatic Position Computation
- *Average Calculation
- *Printable Report Cards (PDF)
- *Dashboard Statistics
- *Responsive Design

8. Technologies to be Used

Frontend

- *React.js
- *JavaScript

Backend

- *Supabase

Database

- *PostgreSQL (Supabase)

*Authentication

- *Supabase Authentication

*Deployment

*Version Control

- (Git & GitHub)

9. Methodology

The project will follow the Agile Software Development methodology.

The development process includes:

1. Requirements gathering
2. System analysis
3. Database design
4. UI/UX design
5. Frontend development
6. Backend API development
7. Database integration
8. Testing
9. Deployment
10. Documentation

10. Expected* *Deliverables

At the end of the project, the team will deliver:

***Fully functional School Results Portal**

***Responsive Web Interface**

***Database**

***RESTful API**

***Technical Documentation**

***User Manual**

***Source Code Repository (GitHub)**

11. Benefits of the System

***The proposed system will:**

- *Reduce errors in result computation
- *Improve processing speed
- *Enhance data security
- *Provide easy access to results
- *Improve record management
- *Support informed academic decision-making
- *Reduce paperwork

12. Future Improvements

*Possible future enhancements include:

- *SMS and email result notifications
- *Parent Portal
- *Mobile Application
- *Fee Management
- *Attendance Tracking
- *AI-powered performance prediction
- *Student transcript generation

13. Conclusion

The School Results Portal will provide an efficient, secure, and reliable solution for managing academic records in educational institutions. By automating result processing and providing secure online access, the system will improve efficiency, reduce human errors, and enhance the overall academic management process.

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